Gravitation $w_{1+\infty}$ Symmetry

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Abstract

Extremely rough notes on gravitation $w_{1+\infty}$ symmetry. Maybe these will turn into something coherent, maybe not!

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1 BMS Invariance of Gravitational Scattering

1.1 The Phase Space of Interest

Any 3+1-dimensional spacetime can be locally put into Bondi gauge

$$ds^2 = -Udu^2 - 2e^{2\beta}dudr + g_{AB}(dx^A - U^A du)(dx^B - U^B du) \tag{1}$$

subject to the areal distance condition

$$\partial_r \det\left(\frac{g_{AB}}{r^2}\right) = 0. \tag{2}$$

An asymptotically flat spacetime can, near future null infinity, be more strongly put into coordinates such that

$$\begin{aligned} ds^2 = &-du^2 - 2dudr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} \\ &+ \frac{2m_B}{r}du^2 + rC_{zz}dz^2 + rC_{\bar{z}\bar{z}}d\bar{z}^2 - 2U_z dudz - 2U_{\bar{z}}dud\bar{z} + \cdots . \end{aligned} \tag{3}$$

If we rewrite the tranverse / angular coordinates $(z, \bar{z})$ as (x^A), we have equivalently

$$\begin{aligned} ds^2 = &\left(-1 + \frac{2m_B(u,z,\bar{z})}{r} + \mathcal{O}\left(\frac{1}{r^2}\right)\right)du^2 - 2\left(1 + \mathcal{O}\left(\frac{1}{r^2}\right)\right)dudr \\ &+ \left(r^2 q_{AB}(u,x) + rC_{AB}(u,x) + \mathcal{O}(1)\right)dx^A dx^B \\ &- 2\left(U_z(u,z,\bar{z}) + \mathcal{O}\left(\frac{1}{r}\right)\right)dudz + \text{c.c.} \end{aligned} \tag{4}$$

Here q_{AB} is the leading-order transverse metric

$$2r^2\gamma_{z\bar{z}}dzd\bar{z} = q_{AB}dx^A dx^B \tag{5}$$

and the areal distance condition forces the shear C_{AB} to be traceless w.r.t. this metric[1]

$$q_{AB}C^{AB} = 0 \tag{6}$$

which is exactly why we have no $C_{z\bar{z}}$ terms. Observe that this is very nearly the statement that the components of the metric are the flat-space Bondi components plus the most general polynomial decay; the only strengthenings are the traceless shear condition coming from the areal distance gauge choice, plus the slightly strong fall-off on g_{ur} which turns out to come from insting on finite energy flux[2] through future null infinity $\mathcal{I}^+$. In fact we will need to make further restrictions, namely

$$U_z = -\frac{1}{2}D^z C_{zz} \tag{7}$$

and the conjugated equation. These come from insisting that the Weyl tensor decay sufficiently quickly near $\mathcal{I}^+$, which is necessary for a conformal compactification. In fact this condition give a tower of additional constraints at further subleading order in r, but we won't need to examine those for this work.

The above discussion has a few subtle quantifiers, so we phrase our phase space of interest carefully. First, we assume we have some reasonable formal definition of future null infinity $\mathcal{I}^+$, which should roughly be "where null rays eventually end up, if they don't hit a singularity or get trapped first." We assume that $\mathcal{I}^+ \simeq S^2 \times \mathbb{R}$, as is the case for flat Minkowski. A spacetime is then asymptotically flat if there exist Bondi-gauge coordinates $(u, r, z, \bar{z})$ near $\mathcal{I}^+$ such that the the metric falls off as described above[3], plus a corresponding set of coordinates near past null infinity $\mathcal{I}^-$ with the analogous properties. Our phase space is exactly the set of asymptotically flat spacetimes in 3+1 dimensions.

1.2 Supertranslations

We are interested in diffeomorphisms which preserve the form of the metric (3). As a side note, it is nearly universal in physics, when discussing a map in coordinates of the form $x \mapsto x'$, to fail to distinguish between a mere change of coordinates and a diffeomorphism. Even more unfortunately, it is common especially in classical literature to always speak of coordinate changes. This is wrong! A coordinate change is always a trivial thing to do, while a diffeomorphism moves points around in spacetime. Here we truly are goin to perform diffeomorphisms while maintaining a fixed coordinate system, so that $x'(x)$ means "the coordinates of the image of the point which had coordinates x. This distinguish doesn't matter if we just want to shut up and calculate, but we do not.

Now as usual in physics, we simplify by studying infinitesimal diffeomorphisms, or equivalently pushforwards under vector fields $\xi = \xi^a \partial_a$ w.r.t. whose components we linearize. For now we want ξ to have at most $\mathcal{O}(1)$ norm near $\mathcal{I}^+$ (we want these fields to generalize translations; soon we will relax this condition to get a generalization of rotations and boosts). So we will require

$$\xi^u, \xi^r \in \mathcal{O}(1), \quad \xi^z, \xi^{\bar{z}} \in \mathcal{O}\left(\frac{1}{r}\right) \tag{8}$$

or more explicitly

$$\begin{aligned} \xi^u(u,r,z,\bar{z}) &= \sum_{n=0}^{\infty} \frac{\xi^{r(n)}(u,z,\bar{z})}{r^n}, \quad \xi^r(u,r,z,\bar{z}) = \sum_{n=0}^{\infty} \frac{\xi^{u(n)}(u,z,\bar{z})}{r^n}, \\ \xi^z(u,r,z,\bar{z}) &= \sum_{n=1}^{\infty} \frac{\xi^{z(n)}(u,z,\bar{z})}{r^n}, \quad \xi^{\bar{z}}(u,r,z,\bar{z}) = \sum_{n=1}^{\infty} \frac{\xi^{\bar{z}(n)}(u,z,\bar{z})}{r^n} \end{aligned} \tag{9}$$

In general, the variation of the metric (or any other same-rank tensor) under an infinitesimal diffeomorphism V is given by

$$\delta_V g_{ab} = \mathcal{L}_V g_{ab} = V^c \partial_c g_{ab} + g_{ac}\partial_b V^c + g_{bc}\partial_a V^c. \tag{10}$$

[1] Plus an infinite tower of additional conditions at subleading order in r

[2] Is this the right observable?

[3] Plus further constraints at subleading orders which, again, we won't need here.

For a metric of the form considered here, we have for example

$$\begin{aligned}\delta_\xi g_{uu} &= \xi^c \partial_c g_{uu} + 2 g_{uc} \partial_u \xi^c \\ &= \left(\xi^r \partial_r + \xi^u \partial_u + \xi^z \partial_z + \xi^{\bar{z}} \partial_{\bar{z}}\right) g_{uu} \\ &\quad + 2 g_{uu} \partial_u \xi^u + 2 g_{ur} \partial_u \xi^r + 2 g_{uz} \partial_u \xi^z + 2 g_{u\bar{z}} \partial_u \xi^{\bar{z}} \\ &= 2 g_{uu} \partial_u \xi^u + 2 g_{ur} \partial_u \xi^r + \mathcal{O}\left(\frac{1}{r}\right) \\ &= -2 \partial_u \xi^u + -2 \partial_u \xi^r + \mathcal{O}\left(\frac{1}{r}\right).\end{aligned} \tag{11}$$

Here we have assumed the pre-variation metric g_{ab} has the form considered above. Similarly,

$$\begin{aligned}\delta_\xi g_{uz} &= \xi^c \partial_c g_{uz} + g_{uc} \partial_z \xi^c + g_{zc} \partial_u \xi^c \\ &= \xi^u \partial_u g_{uz} + g_{uu} \partial_z \xi^u + g_{ur} \partial_z \xi^r + g_{zu} \partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right) \\ &= -\xi^u \partial_u U_z - \partial_z \xi^u - \partial_z \xi^r - U_z \partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right)\end{aligned} \tag{12}$$

Further similar calculations give

$$\begin{aligned}\mathcal{L}_\xi g_{ur} &= -\partial_u \xi^u + \mathcal{O}\left(\frac{1}{r}\right), \\ \mathcal{L}_\xi g_{zr} &= r^2 \gamma_{z\bar{z}} \partial_r \xi^{\bar{z}} - \partial_z \xi^u + \mathcal{O}\left(\frac{1}{r}\right), \\ \mathcal{L}_\xi g_{z\bar{z}} &= r \gamma_{z\bar{z}} \left[2\xi^r + r D_z \xi^z + r D_{\bar{z}} \xi^{\bar{z}}\right] + \mathcal{O}(1).\end{aligned} \tag{13}$$

Less interestingly,

$$\delta_\xi g_{rr} = \mathcal{O}\left(\frac{1}{r}\right). \tag{14}$$

We want to determine which vector fields ξ respect the fall-offs and the Bondi gauge conditions. For example, maintaining the condition

$$g_{ur} = -1 + \mathcal{O}\left(\frac{1}{r}\right) \tag{15}$$

forces

$$-\partial_u \xi^u = \mathcal{O}\left(\frac{1}{r}\right) \tag{16}$$

hence

$$\xi^u(u, r, z, \bar{z}) = f(z, \bar{z}) + \mathcal{O}\left(\frac{1}{r}\right) \tag{17}$$

for some function f on S^2. Applying the result of the conditions (and carefully tracking Christofells in manipulating the sphere-covariant derivatives $D_z, D_{\bar{z}}$) gives us the general solution

$$\xi = \xi^a \partial_a = f \xi_u - \frac{1}{r}\left(D^z f \partial_z + D^{\bar{z}} f \partial_{\bar{z}}\right) + D^z D_z f \partial_r + (\text{componentwise-subleading terms}) \tag{18}$$

2 $w_{1+\infty}$ and the Celestial Sphere

Scratch notes following [**?**].

2.1 Conformally Soft Gravitons

Denote by $G^+_\Delta(z,\bar{z})$ the celestial dual to a positive-helicity[4] bulk graviton. The authors of [**?**] find by analyzing bulk collinear limits that two such operators have OPE

$$G^+_{\Delta_1}(z_1,\bar{z}_1)G^+_{\Delta_2}(z_2,\bar{z}_2) = -\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^{\infty} B(\Delta_1-1+n,\Delta_2-1)\frac{\bar{z}_{12}^{n+1}}{n!}\bar{\partial}^n G^+_{\Delta_1+\Delta_2}(z_2,\bar{z}_2)+\mathcal{O}\left(z_{12}^0\right). \tag{19}$$

Here B is the Beta function

$$B(x,y)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \tag{20}$$

holomorphic poles[5]. These operators have conformal spin $s=+2$ and scaling weights Δ, hence conformal weights

$$\left(h,\bar{h}\right)=\left(\frac{\Delta+s}{2},\frac{\Delta-s}{2}\right)=\left(\frac{\Delta}{2}+1,\frac{\Delta}{2}-1\right). \tag{21}$$

These operators are, by definition, the result of Mellin-transformating a bulk scattering amplitude of gravitons, necessarily in an asymptotically flat spacetime, to produce a boundary correlation function.

$$\langle G^\pm_\Delta(z,\bar{z})\cdots\rangle \equiv \int_0^\infty \frac{d\omega}{\omega}\omega^\Delta\cdots\mathcal{A}\left(\omega\hat{p}(z,\bar{z}),\cdots\right). \tag{22}$$

From the well-known property of Mellin transformations

$$\int_0^\infty \frac{d\omega}{\omega}\omega^\Delta\left(\lim_{\omega\to 0}\omega f(w)\right)=\lim_{\Delta\to 1}(\Delta-1)\int_0^\infty\frac{d\omega}{\omega}\omega^\Delta\omega f(w) \tag{23}$$

we see that a bulk amplitude involving a soft graviton

$$\lim_{\omega\to 0}\omega\mathcal{A}\left(\omega\hat{p}(z,\bar{z}),\cdots\right) \tag{24}$$

corresponds on the boundary to taking the limit

$$\lim_{\Delta\to 1}(\Delta-1)G^+_\Delta(z,\bar{z}). \tag{25}$$

This nicely agrees with (19), which we see has an order-one pole at $\Delta_i=1$, rendering this limit finite. In fact we see it has a tower of order-one poles at $\Delta_i=1,0,-1,\ldots$. We therefore define more generally

$$H^k(z,\bar{z})\equiv\lim_{\varepsilon\to 0}\varepsilon G^+_{k+\varepsilon}(z,\bar{z}),\ k\in\mathbb{Z}. \tag{26}$$

We will shortly find that these live in a shortened representation, so that only a subset of k values give a nonzero result.

These objects turn out to generate an interesting algebra. We first expand in the antiholomorphic coordinate

$$H^k(z,\bar{z})=\sum_n\frac{H^k_n(z)}{\bar{z}^{n+\frac{k-2}{2}}} \tag{27}$$

where the shift in the powers of $\bar{z}$ is required by H^k having a nonzero antiholomorphic conformal weight. Again, we will find that the relevant values of n are limited. We now interpret the objects $H^k_n(z)$ as holomorphic currents on the boundary. It will be illustrative to note the equivalent integral representation

$$H^k_n(z)=\oint_0\frac{d\bar{z}}{2\pi i}\bar{z}^{n+\frac{k}{2}-2}H^k(z,\bar{z}). \tag{28}$$

[4] TODO: Comment on choice to restrict to positive-helicity sector.

[5] TODO: Comment more on this choice.

2.2 Shortened Representations

To understand how these currents work, we compute their OPEs against celestial gravitons. Using the results of the last section, we find

$$\begin{aligned} H_n^k(z_1)G_\Delta^+(z_2,\bar{z}_2) &= \oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}H^k(z_1,\bar{z}_1)G_\Delta^+(z_2,\bar{z}_2) \\ &= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}G_{k+\varepsilon}^+(z_1,\bar{z}_1)G_\Delta^+(z_2,\bar{z}_2) \\ &= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\left(-\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^{\infty}B(k+\epsilon-1+n,\Delta-1)\frac{\bar{z}_{12}^{n+1}}{n!}\bar{\partial}^n G_{k+\epsilon+\Delta}^+(z_2,\bar{z}_2)+\mathcal{O}\left(z_{12}^0\right)\right) \\ &= -\frac{\kappa}{2}\frac{1}{z_{12}}\lim_{\varepsilon\to 0}\varepsilon\sum_{n=0}^{\infty}\frac{1}{n!}B(k+\epsilon-1+n,\Delta-1)\bar{\partial}^n G_{k+\epsilon+\Delta}^+(z_2,\bar{z}_2)\oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\bar{z}_{12}^{n+1}+\mathcal{O}\left(z_{12}^0\right). \end{aligned} \tag{29}$$

The innermost integral can be evaluated by binomial expansion as

$$\begin{aligned} \oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}(\bar{z}_1-\bar{z}_2)^{n+1} &= \sum_{\ell=0}^{n+1}\binom{n+1}{\ell}(-\bar{z}_2)^{n+1-\ell}\underbrace{\oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\bar{z}_1^{\ell}}_{=\delta_{n+\ell+\frac{k}{2}-2,0}} \\ &= \sum_{\ell=0}^{n+1}\frac{(n+1)!}{\ell!(n+1-\ell)!}(-\bar{z}_2)^{n+1-\ell}\delta_{n+\ell+\frac{k}{2},2}. \end{aligned} \tag{30}$$

For k, n such that this is ever nonzero, the only contributing term in the sum is

$$\ell = 2-n-\frac{k}{2} \implies \binom{n+1}{\ell} = \frac{(n+1)!}{\left(2-n-\frac{k}{2}\right)!\left(2n+\frac{k}{2}-1\right)} \tag{31}$$